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#This installs the biostrings and decipher from bioconductor

BiocManager::install("Biostrings")
BiocManager::install("DECIPHER")

# Load required libraries
library(Biostrings)
library(DECIPHER)


#Reads in megahit assembly results for each sample

allcontigs <- c()
for(i in 1:6){
contigs <- readDNASTringSet(paste0('C:/Users/laine/Downloads/forlaine/t', i,
'_out/final.contigs.fa'))
allcontigs <- c(allcontigs, contigs)
}

# This loop outputs a list, so we flatten the list
allcontigs <- do.call(c, allcontigs)

# There are some small fragments of genomes in the assmebly; we are only going to align the
#biggest parts (aka the ones that are bigger than 5 kbp; the genome itself is around 18 kbp).
toalign <- allcontigs[which(nchar(allcontigs) > 5000)]
names(toalign) <- 1:length(toalign)


#Display all imported contigs

>allcontigs

# Perform a multiple sequence alignment on the selected contigs
> Alignment<-AlignSeqs(toalign)


#Open an interactive HTML file to visualize and browse the sequence alignment

BrowseSeqs(Alignment,htmlFile = "final_project.html")


#Generate and display a maximum likelihood phylogenetic tree based on the aligned sequences
> Tree<-Treeline(Alignment,method = "ML",showPlot = TRUE)

```